Test Report - TIA-Demo Prototype-002

Test Information

Project:	TIA-Demo Simulation
Operator:	Tobias
Date:	18.10.2026
Time:	18:50:52
Duration:	19,15s

Device Under Test

Serial Number:	Prototype-002
Device Name:	TIA-Demo
Revision:	B
Variant:	I2C DAC + MCU
Part Number:	-

Additional Notes:
DIP switches determine which tests are executed

FAIL

Step	Name	Lower Limit	Measured	Upper Limit	Result
Setup					
2	VIN 12V?	10,8V	12,031V	13,2V	✔
3	U1 VOUT 3V3?	2,97V	3,302V	3,63V	✔
4	D2 Forward Voltage	1,71V	1,903V	2,09V	✔
5	R1 Voltage Drop	1,17V	1,332V	1,43V	✔
6	D2 Forward Current	1,17mA	1,332mA	1,43mA	✔
10	Segment A ON?	2,97V	3,293V	3,63V	✔
12	Segment B ON?	2,97V	3,275V	3,63V	✔
14	Segment C ON?	2,97V	3,265V	3,63V	✔
16	Segment D ON?	2,97V	3,273V	3,63V	✔
18	Segment E ON?	2,97V	3,286V	3,63V	✔
20	Segment F ON?	2,97V	3,299V	3,63V	✔
22	Segment G ON?	2,97V	3,264V	3,63V	✔
23	### Read DIP Switches ###	0	0	255	✔
LED Test					
1	### DIP Switch 1 ON? ###	0	0	0	✔
4	D3 Red Forward Voltage	1,53V	1,697V	1,87V	✔
5	R22 Voltage Drop	1,35V	1,456V	1,65V	✔
6	D3 Red Forward Current	1,143mA	1,213mA	1,397mA	✔
8	D3 Green Forward Voltage	2,07V	2,285V	2,53V	✔

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9	R23 Voltage Drop	900mV	863mV	1,1V	✗
10	D3 Green Forward Current	747uA	719,167uA	913uA	✗
12	D3 Blue Forward Voltage	2,25V	2,531V	2,75V	✓
13	R24 Voltage Drop	675mV	752mV	825mV	✓
14	D3 Green Forward Current	576uA	626,667uA	704uA	✓
15	RGB LED User Response	1	1	1	✓
Button Test					
1	### DIP Switch 2 ON? ###	0	0	0	✓
3	Button 1 High?	2,97V	3,252V	3,63V	✓
4	Button 1 Low?	-200mV	56mV	200mV	✓
5	Button 2 High?	2,97V	3,287V	3,63V	✓
6	Button 1 Low?	-200mV	49mV	200mV	✓
Voltage Divider Test					
1	### DIP Switch 3 ON? ###	0	0	0	✓
3	Divider with 680R	189mV	220mV	231mV	✓
4	Divider with 1k2	315mV	368mV	385mV	✓
5	Divider with 2k2	531mV	630mV	649mV	✓
6	Divider with 3k9	828mV	990mV	1,012V	✓
7	Divider with 6k8	1,17V	1,23V	1,43V	✓
Adjust Potentiometer					
1	### DIP Switch 4 ON? ###	0	0	0	✓
Temperature Sensor					
1	### DIP Switch 5 ON? ###	0	0	0	✓
3	Temperature Sensor VOUT	693mV	740mV	847mV	✓
4	Temperature	22,5°C	24°C	27,5°C	✓
I2C DAC					
1	### DIP Switch 6 ON? ###	0	0	0	✓
3	Set I2C DAC to 0.5V	1	1	1	✓
4	Measure DAC Output 0.5V?	450mV	504mV	550mV	✓
5	Set I2C DAC to 1.0V	1	1	1	✓
6	Measure DAC Output 1.0V?	900mV	1,024V	1,1V	✓

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Step	Name	Lower Limit	Measured	Upper Limit	Result
7	Set I2C DAC to 1.5V	1	1	1	✔
8	Measure DAC Output 1.5V?	1,35V	1,532V	1,65V	✔
9	Set I2C DAC to 2.0V	1	1	1	✔
10	Measure DAC Output 2.0V?	1,8V	2,065V	2,2V	✔
11	Set I2C DAC to 2.5V	1	1	1	✔
12	Measure DAC Output 2.5V?	2,25V	2,515V	2,75V	✔
13	Set I2C DAC to 3.0V	1	1	1	✔
14	Measure DAC Output 3.0V?	2,7V	3,121V	3,3V	✔
Program MCU					
1	### DIP Switch 7 ON? ###	0	0	0	✔
3	Program MCU	0	0	0	✔
4	Get FW-Version	1,2	1,2	1,2	✔
Test MCU					
1	### DIP Switch 8 ON? ###	0	0	0	✔
3	Read POT2 Value from MCU	1350mV	1502mV	1650mV	✔
4	Configure MCU DAC to 0.5V	1	1	1	✔
5	Read DAC Output from MCU ADC	450mV	505mV	550mV	✔
6	Measure DAC Output Voltage	450mV	507mV	550mV	✔
7	Configure MCU DAC to 1.0V	1	1	1	✔
8	Read DAC Output from MCU ADC	900mV	1005mV	1100mV	✔
9	Measure DAC Output Voltage	900mV	1,002V	1,1V	✔
10	Configure MCU DAC to 1.5V	1	1	1	✔
11	Read DAC Output from MCU ADC	1350mV	1510mV	1650mV	✔
12	Measure DAC Output Voltage	1,35V	1,514V	1,65V	✔
13	Configure MCU DAC to 2.0V	1	1	1	✔
14	Read DAC Output from MCU ADC	1800mV	2015mV	2200mV	✔
15	Measure DAC Output Voltage	1,8V	2,005V	2,2V	✔
16	Configure MCU DAC to 2.5V	1	1	1	✔
17	Read DAC Output from MCU ADC	2250mV	2507mV	2750mV	✔
18	Measure DAC Output Voltage	2,25V	2,509V	2,75V	✔
19	Configure MCU DAC to 3.0V	1	1	1	✔
20	Read DAC Output from MCU ADC	2700mV	3020mV	3300mV	✔

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Step	Name	Lower Limit	Measured	Upper Limit	Result
21	Measure DAC Output Voltage	2,7V	3,006V	3,3V	✔
22	Measure Timer Frequency	4,5Hz	4,89Hz	5,5Hz	✔

Used Test Equipment

Test Hardware:

Device Name: **compact Test Interface Adapter**

Serial Number: **2026001**

Firmware Version: **0.5.3**

Test Equipment:

Rigol DM858: -